

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE Supplementary Examination – Winter 2023 Course: B. Tech. Branch : Mechanical Engineering/ Mechanical Engineering (Sandwich) Semester :VI Subject Code & Name: BTMEC604B_Y19 IC Engine Max Marks: 60 Date:23/01/2024 Duration: 3 Hr.			
Instructions to the Students: 1. All the questions are compulsory. 2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question. 3. Use of non-programmable scientific calculators is allowed. 4. Assume suitable data wherever necessary and mention it clearly.			
		(Level/Co)	Marks
Q. 1	Solve Any Two of the following.		12
	a) Explain the following with neat sketch: (a) Connecting Rod (b) Cam Shaft (c) Valves (d) Crank Shaft	Remembering	06
	b) Discuss about the working of a 4 stroke petrol engine with neat sketch.	Understanding	06
	c) Differentiate between 4 stroke and 2 stroke Diesel engine.	Analyzing	06
Q. 2	Solve Any Two of the following.		12
	a) Explain the phenomenon of Knock in Compression Ignition Engine.	Remembering	06
	b) State the various factors which affect combustion in CI engine	Understanding	06
	c) With neat sketch explain the phenomenon of combustion in SI engine.		06
Q. 3	Solve Any Two of the following.		12
	a) Explain the battery ignition system of engine with a neat sketch.	Remembering	06
	b) Compare the merits and demerits of air and water cooling systems.	Evaluating	06
	c) Explain Simple Carburetor with a neat sketch	Understanding	06
Q. 4	Solve Any Two of the following.		12
	a) How Supercharging improves the power output of engine? Explain any one type of supercharger with neat sketch.	Analyzing	06
	b) Explain Heat balance sheet in IC engine and its procedure to calculate.	Evaluating	06
	c) State different exhaust emissions of I.C. Engine. Explain its effects on ecology.	Evaluating & Analyzing	06

Q. 5	Solve Any Two of the following.		12
	a) Explain types of fuel cells and what are challenges that fuel cells are facing?	Understanding	06
	b) What are the advantages and disadvantages of hybrid vehicles	Remembering	06
	c) What are the functions of Electronic control systems in IC Engine.	Analyzing	06
	*** End ***		